ModelIt!

Teacher's Guide

AI, Robotics & Ethics — Modeling Hard Choices

When Technology Meets Nature, Who Decides What's Right?

Quick Reference

Grade Level	Grade NE
NGSS Standard	HS-ETS1-1, HS-ETS1-3, HS-LS2-7, HS-LS4-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ETS1-1, HS-ETS1-3, HS-LS2-7, HS-LS4-6

Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints. Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity. Create or revise a simulation to test a solution to mitigate adverse impacts of human activity on biodiversity.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students use 9-component model outputs as quantitative evidence to argue for or against specific AI conservation deployment strategies across multiple ethical dimensions.
Disciplinary Core Idea	**ETS1.B Developing Possible Solutions** — Students evaluate technology solutions by weighing ecological benefit criteria against ethical constraints including transparency, animal autonomy, and public trust.
Crosscutting Concept	**Cause and Effect** — Students identify that technology deployment creates cascading and sometimes contradictory effects — the same input (high capability) simultaneously causes positive outcomes (ecological benefit) and negative outcomes (unintended consequences), requiring systems-level analysis.

CAST Testing Alignment

MULTIPLE CHOICE:

A conservation agency deploys an autonomous drone network to monitor endangered sea turtles. The drones operate with high capability but low transparency. According to the model, which outcome is MOST LIKELY? A) High public trust due to ecological benefit B) Low public trust despite ecological benefit C) High ethical justification due to crisis severity D) Low unintended consequences due to regulation

CONSTRUCTED RESPONSE:

A government proposes using AI-controlled robotic predators to manage an invasive species threatening native birds. Using your 9-component model, evaluate this proposal by analyzing Technology Capability, Transparency, Crisis Severity, and Regulation. Then argue whether this deployment is ethically justified, addressing ecological benefit, unintended consequences, animal autonomy, and public trust.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson builds on Level 1's simple three-factor ethics model by expanding to a full 9-component system that captures the real complexity of deploying AI and robotics in conservation. Students now grapple with feedback loops, competing values, and the uncomfortable reality that there is no deployment strategy that maximizes all ethical outcomes simultaneously.

AI in Conservation Today

AI is already transforming conservation: machine learning identifies species from camera trap images, drones monitor deforestation in real time, acoustic sensors detect poaching activity, and predictive algorithms guide ranger patrol routes. These technologies have produced measurable gains — but each deployment raises questions about surveillance, data ownership, community consent, and what happens when algorithms make mistakes.

The Capability-Consequence Paradox

More powerful conservation technology produces greater ecological benefits — but also greater potential for harm. A basic camera trap has limited benefit but also limited risk. A fully autonomous system that can track, identify, and intervene with wildlife has enormous potential benefit but also enormous potential for unintended consequences: disrupting migration patterns, habituating animals to technology, creating dependency, or misidentifying threats. This paradox means that the most powerful tools demand the most careful governance.

Transparency and Trust

Public trust in conservation technology is fundamentally tied to transparency. When communities understand how a technology works, what data it collects, and who controls the decisions, they are far more likely to support its deployment. When technology operates opaquely — collecting data without explanation, making decisions without accountability — public opposition grows regardless of ecological results. The history of conservation is filled with well-intentioned projects that failed because they lost community trust.

The Regulation Dilemma

Regulation protects against harm but also limits benefit. Strict rules slow deployment, increase costs, and may prevent timely intervention during ecological emergencies. No regulation allows rapid action but invites recklessness, profiteering, and abuse. The challenge is finding proportional regulation — strict enough to prevent harm, flexible enough to allow benefit, and adaptive enough to evolve as technology changes. Our model reveals this tension directly: increasing Regulation Strictness decreases Unintended Consequences but also decreases Ecological Benefit.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Technology Capability Level, Transparency Level, Ecological Crisis Severity, and Regulation Strictness as external components (conditions and policy choices we can set). Ecological Benefit, Unintended Consequences, Animal Autonomy Impact, Public Trust, and Ethical Justification Score are internal components (outcomes produced by the system).

E **Establish Relationships**

Students discover complex and sometimes contradictory relationships: Technology Capability positively affects BOTH Ecological Benefit AND Unintended Consequences. Transparency positively affects Public Trust. Unintended Consequences negatively affect Public Trust (feedback loop). Crisis Severity positively affects Ethical Justification. Animal Autonomy Impact negatively affects Ethical Justification. Regulation reduces Unintended Consequences but also reduces Ecological Benefit. Low Public Trust drives increased Regulation (feedback loop).

V **Visualize & Model**

Students build the full 9-component model with four external inputs feeding five internal outputs through interconnected, sometimes contradictory relationships including feedback loops.

E **Evaluate Outcomes**

Students run four scenarios — Transparent and Regulated, Unregulated Cowboy, Emergency Override, and Surveillance State — to test how different governance philosophies produce different ethical outcomes.

R **Revise & Extend**

Students add Community Benefit, Data Security, or Long-Term Dependency to see how additional real-world factors further complicate the ethical landscape.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In Level 1, we asked 'should we or should we not?' Today we ask something harder: WHEN we use AI and robots to protect nature, who decides what is ethical — and how?"*

Watch For

Show a montage of real conservation technology: camera traps, drones, acoustic sensors, robotic fish. Ask: What could go wrong with each?

Slide 2: Learning Objectives

3 min

Say: *"We are building a 9-component model today — the most complex system in this series. Nine moving parts, feedback loops, and no easy answers."*

Watch For

Pre-teach 'algorithmic transparency,' 'autonomous system,' and 'animal autonomy.' Connect to Level 1 vocabulary: intervention, risk assessment, unintended consequences.

Slide 3: Big Question

3 min

Say: *"An AI drone detects a poacher and decides to alert rangers. Another AI decides to herd endangered animals away from a wildfire. A third AI decides which habitat patches to restore first. Who programmed those decisions? Who approved them? Who is accountable if something goes wrong?"*

Watch For

Think-pair-share: If an AI makes a mistake that harms wildlife, who is responsible — the engineer, the government, or the AI itself? Collect diverse perspectives.

Slide 4: LEVER Framework

3 min

Say: *"This model has something new — FEEDBACK LOOPS. When unintended consequences happen, public trust drops, which causes MORE regulation, which reduces ecological benefit. Actions have echoes."*

Watch For

Draw the feedback loop on the board: Unintended Consequences → Public Trust drops → Regulation increases → Ecological Benefit drops. Ask: How do you break this cycle?

Slide 5: Activity 1: Components

10 min

Say: *"Four things we can set: how powerful the tech is, how transparent we are, how severe the crisis is, and how strict the rules are. Five things the system produces. Sort them."*

Watch For

Guide component sorting. Emphasize: this is the first model with NINE components. The externals are policy and design CHOICES. The internals are CONSEQUENCES of those choices.

Slide 6: Activity 2: Connections

12 min

Say: *"Here is where it gets intense. Technology Capability feeds Ecological Benefit with a positive arrow AND feeds Unintended Consequences with ANOTHER positive arrow. The same input causes good AND bad. That is the paradox."*

Watch For

Build relationships step by step. Key moments: (1) dual arrows from Tech Capability, (2) Transparency to Public Trust positive, (3) Unintended Consequences to Public Trust negative, (4) feedback loop from Public Trust to Regulation. Have students trace each path.

Slide 7: Activity 3: Simulation

14 min

Say: *"Four governance philosophies. Transparent and Regulated, Unregulated Cowboy, Emergency Override, and Surveillance State. Predict BEFORE we run. Which approach scores highest on ethical justification?"*

Watch For

Run all four scenarios. After each, pause for comparison. Key discussion: Emergency Override — does crisis severity change what is ethical? Surveillance State — can regulation substitute for transparency?

Slide 8: Discoveries

5 min

Say: *"No deployment strategy maximizes everything. But transparency combined with proportional regulation consistently produces the most ethically defensible outcomes. The model does not tell us what is right — it shows us what we are trading away with every choice."*

Watch For

Summarize key findings. Revisit the big question: Who decides? Discuss: Should it be engineers, communities, governments, or all three? What role does the model play in that decision?

Slide 9: STEM Challenge

12 min

Say: *"Now you become conservation technology designers AND ethicists. Design a fictional technology, evaluate it through your model, and pitch it to the class."*

Watch For

Distribute project materials. Teams choose a crisis, design a technology, create a blueprint and ethical impact assessment, draft a public communication plan, and prepare a 2-minute pitch.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Technology Capability Level, Transparency Level, Ecological Crisis Severity, Regulation Strictness
- **INTERNAL:** Ecological Benefit, Unintended Consequences, Animal Autonomy Impact, Public Trust, Ethical Justification Score

Relationships

- **Technology Capability to Ecological Benefit:** POSITIVE (+) — More powerful AI and robotic systems can monitor more area, respond faster, and manage more complex ecological interventions. Higher capability translates directly to greater conservation impact.
- **Technology Capability to Unintended Consequences:** POSITIVE (+) — More powerful systems interact with ecosystems in more ways, creating more opportunities for unexpected outcomes — algorithm errors, behavioral disruption, cascading ecological effects that simpler tools would not trigger.
- **Technology Capability to Animal Autonomy Impact:** POSITIVE (+) — More capable systems interact more directly with wildlife — tracking, herding, deterring, or managing animal behavior. Higher capability means more intrusion into natural behavioral patterns.
- **Transparency Level to Public Trust:** POSITIVE (+) — When the public can see how the technology works, what data it collects, and how decisions are made, trust increases substantially. Transparency is the single strongest predictor of sustained public support.
- **Unintended Consequences to Public Trust:** NEGATIVE (-) — Every unintended negative outcome erodes public confidence. Even one highly visible failure can destroy years of trust-building, regardless of the overall ecological benefit achieved.
- **Ecological Crisis Severity to Ethical Justification Score:** POSITIVE (+) — The more severe the crisis, the more ethically acceptable it becomes to deploy powerful technology — emergency conditions shift the moral calculus toward action even when risks are elevated.
- **Animal Autonomy Impact to Ethical Justification Score:** NEGATIVE (-) — Greater disruption to natural animal behavior weakens ethical justification. Wildlife has intrinsic value as autonomous beings — the more technology controls or constrains that autonomy, the harder the intervention is to justify morally.
- **Regulation Strictness to Unintended Consequences:** NEGATIVE (-) — Stricter regulations require more testing, more safeguards, and more oversight, which reduces the likelihood and severity of unintended negative outcomes.
- **Regulation Strictness to Ecological Benefit:** NEGATIVE (-) — Stricter regulations slow deployment, limit the scope of intervention, and increase costs — all of which reduce the total ecological benefit that can be achieved within a given timeframe.
- **Public Trust to Regulation Strictness:** NEGATIVE (-) FEEDBACK LOOP — When public trust drops, political pressure increases for stricter regulations. Low trust drives regulatory backlash, which further constrains the technology. This feedback loop can spiral: unintended consequences → trust drops → more regulation → less benefit → question the whole program.

Key Discoveries

- Technology capability is a double-edged sword — more powerful systems produce greater ecological benefits AND greater unintended consequences simultaneously
- Transparency is the single strongest driver of public trust — without it, even beneficial technology faces public opposition
- Unintended consequences erode public trust through a feedback loop: low trust leads to stricter regulation, which reduces ecological benefit
- Crisis severity shifts the ethical calculus — actions that are unjustifiable in normal times may become necessary during emergencies
- Animal autonomy impact works AGAINST ethical justification — the more technology controls animal behavior, the harder it is to justify morally
- Regulation reduces unintended consequences but ALSO reduces ecological benefit — there is no regulation level that maximizes both

- The highest ethical justification scores require BOTH high transparency AND moderate regulation — neither alone is sufficient

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"More advanced technology is always better for conservation"	The model directly contradicts this: higher Technology Capability increases Ecological Benefit AND Unintended Consequences simultaneously. The most powerful tools carry the most risk. A simpler technology with fewer unintended consequences and higher public trust may achieve more long-term conservation success than a powerful system that triggers a regulatory backlash. The right technology is the one proportional to the crisis, not the most advanced one available.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Strict regulation solves all ethical problems"	Regulation reduces unintended consequences, but it also reduces ecological benefit — and it cannot substitute for transparency. The Surveillance State scenario proves this: strict rules with low transparency still produces low public trust and low ethical justification. Regulation without transparency creates a 'trust us, we have rules' dynamic that the public sees through. Effective governance requires BOTH regulatory safeguards AND transparent communication.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If the ecological benefit is large enough, nothing else matters"	This is consequentialist thinking — and the model shows its limits. Even with maximum ecological benefit, high unintended consequences, low public trust, and high animal autonomy impact drag down the ethical justification score. Conservation that ignores community trust, animal welfare, and transparency may achieve short-term gains but creates long-term opposition that undermines the entire program. Sustainable conservation requires ethical legitimacy, not just ecological results.